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SUPPLEMENTARY MATERIAL

The Two Roots of the Tetrad Equation

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Abstract
The tetrad equation $g = e^T \eta\, e$ forces $(\det e)^2 = \det g / \det \eta$, an equation of the form $y^2 = c$. We prove that its scheme has exactly two connected components, one over each root $\det e = \pm\sqrt{c}$, and that neither splits further over $\mathbb{C}$. A change of frame exchanges the roots if and only if it reverses orientation; in three spatial dimensions spatial parity does, and it is the only such map that commutes with every rotation and reflection of an observer's space. For future-pointing frames the two roots are the two tetrahedra of the stellated octahedron. The metric determines $(\det e)^2$ and never $\det e$, so keeping one root, as the restriction to proper orthochronous frames does, is a convention.

Keywords: tetrad, scheme, Lorentz group, root-blindness, spatial parity, stellated octahedron, proof

Setting

At a point of a spacetime with $1 + D$ dimensions and metric g of signature $(-, +, \ldots, +)$, a *tetrad* is a real square matrix $e = (e^a{}_\mu)$ with

$$e^T \eta\, e = g, \qquad \eta = \mathrm{diag}(-1, I_D). \tag{1}$$

The entries of e are the unknowns, and Eq. (1) is a set of quadratic equations in them. They define the affine scheme $X_g = \mathrm{Spec}\,\mathbb{R}[e^a{}_\mu]/(e^T \eta\, e - g)$ over $\mathbb{R}$ [1, 2], whose real points $X_g(\mathbb{R})$ are the real tetrads. The orthogonal group scheme $O(\eta)$, with real points the Lorentz group $O(1, D)$, acts by $e \mapsto \Lambda e$. We write $P = \mathrm{diag}(1, -I_D)$ (spatial parity), $T = \mathrm{diag}(-1, I_D)$ (time reversal) and $R = \mathrm{diag}(1, -1, 1, \ldots, 1)$ (one spatial reflection). Since g and η have the same signature, $c = \det g / \det \eta > 0$.

Two Lemmas

Lemma 1 (time cone). *If $\Lambda \in O(1, D)$ and $\eta(u, u) < 0$, then $\mathrm{sgn}(\Lambda u)^0 = \mathrm{sgn}\,\Lambda^0{}_0 \cdot \mathrm{sgn}\,u^0$, and $\Lambda \mapsto \mathrm{sgn}\,\Lambda^0{}_0$ is a homomorphism.*

Proof. $\Lambda^T \eta \Lambda = \eta$ gives $\Lambda \eta \Lambda^T = \eta$, whose $(0, 0)$ entry is $|r|^2 = (\Lambda^0{}_0)^2 - 1$ for $r = (\Lambda^0{}_i)$. A timelike u has $|\vec{u}| < |u^0|$, so in $(\Lambda u)^0 = \Lambda^0{}_0 u^0 + r \cdot \vec{u}$ the second term is smaller in modulus than the first. Taking $u = \Lambda'_{\cdot 0}$ gives the homomorphism. □

Lemma 2 (centre). *For $D \geq 1$, the elements of $O(1, D)$ commuting with all of $O(1, D)$ are $\pm I$, and those of $O(D)$ commuting with all of $O(D)$ are $\pm I_D$.*

Proof. Write $Z = (\alpha, w^T; v, Y)$. Commuting with $\mathrm{diag}(1, -I_D)$ gives $v = w = 0$; with all diagonal sign matrices and all permutation matrices in $O(D)$, $Y = \lambda I_D$, which also settles $O(D)$. A boost in the 0–1 plane forces $\alpha = \lambda$, and $Z^T \eta Z = \lambda^2 \eta$ gives $\lambda = \pm 1$. □

The Two Roots

Theorem 1 (the scheme). *Let e_0 be a real tetrad. Then:*

(a) *$\Lambda \mapsto \Lambda e_0$ is an isomorphism $O(\eta) \to X_g$. Every real tetrad is Λe_0 for exactly one $\Lambda \in O(1, D)$.*

(b) *On X_g, $(\det e)^2 = c$. So $\det : X_g \to \{y^2 = c\}$, and $X_g = X_+ \sqcup X_-$, where $X_\pm$ is the fibre over $y = \pm\sqrt{c}$. Each $X_\pm$ is connected, and stays connected over $\mathbb{C}$: X_g has exactly two connected components, the two roots.*

(c) *$e \mapsto \Lambda e$ exchanges the roots if and only if $\det \Lambda = -1$. T and R always do; P does if and only if D is odd; $-I$ does if and only if D is even.*

Proof. (a) From Eq. (1), $(\det e)^2 \det \eta = \det g$, a unit, so e is invertible on X_g and $e \mapsto e\, e_0^{-1}$ is the inverse map: it lands in $O(\eta)$ because $e_0^{-T} e^T \eta\, e\, e_0^{-1} = e_0^{-T} g\, e_0^{-1} = \eta$. (b) The identity above divided by $\det \eta$. The target is two points, so $X_\pm$ are open and closed. Under (a), $X_+ \cup X_-$ is $O(\eta) = SO(\eta) \sqcup R\, SO(\eta)$, and $SO(\eta)$ is connected over $\mathbb{C}$, where it is $SO(1 + D, \mathbb{C})$ [3]. (c) $\det(\Lambda e) = \det \Lambda \det e$, with $\det T = \det R = -1$, $\det P = (-1)^D$ and $\det(-I) = (-1)^{D+1}$. □

Theorem 2 (real points and parity). *Fix a future-directed timelike t and set $\tau(e) = \mathrm{sgn}(e\, t)^0$.*

(a) *$\tau(\Lambda e) = \mathrm{sgn}\,\Lambda^0{}_0 \cdot \tau(e)$. The real points of each root form two pieces, $\tau = \pm 1$; $X_g(\mathbb{R})$ has four.*

(b) *Among future-pointing frames, $\tau = +1$, the root $\mathrm{sgn}\det e$ is the orientation of the*

spatial legs. A change of frame that keeps the observer, $\Lambda = \mathrm{diag}(1, Q)$, exchanges the roots if and only if $\det Q = -1$, and for D odd the only such Q that commutes with every rotation and reflection of the observer's space is $-I_D$: spatial parity P.

(c) *Other than the identity, $e \mapsto -e$ is the only change of frame that commutes with all others. For D odd it keeps the root and reverses τ: $-e = T(Pe)$.*

Proof. (a) $u = e\,t$ has $\eta(u,u) = g(t,t) < 0$; apply Lemma 1. Under Theorem 1(a), τ and the root separate the four components $X \cdot SO^+(1,D)$, $X \in \{I, T, R, RT\}$, of $O(1,D)$ (for $D = 3$, see [4]). (b) $\det \mathrm{diag}(1, Q) = \det Q$; the centre of $O(D)$ is $\pm I_D$ (Lemma 2), and $\det(-I_D) = -1$ for D odd. (c) Lemma 2, with $\det(-I) = (-1)^{D+1}$ and $(-I)^0{}_0 = -1$. □

Theorem 3 (root-blindness). *The metric determines $(\det e)^2 = c$ and never $\det e$. Every quantity built from g, from the connection and curvature to the Einstein–Hilbert action, takes one value on both roots, for tetrad fields as at a point. Restricting frames to $SO^+(1,D)\,e_0$, the usual choice for oriented and time-oriented frames [5, 6], keeps one root; g does not determine which.*

Proof. By Theorem 1(a) every point of X_g gives the same g, and $Pe(x)$ solves Eq. (1) wherever $e(x)$ does. $SO^+(1,D) \subset SO(\eta)$, and e_0 may be taken in either root. □

The Two Tetrahedra

Take $D = 3$. Let E_a be the frame vectors of e, with $g(E_a, E_b) = \eta_{ab}$, and W the rest space spanned by E_1, E_2, E_3. Letter 1's directions [7] are $n_a = \sum_i M_{ia} E_i$, with

$$M = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix}, \qquad (2)$$

the regular tetrahedron forced by the tetrad [8].

Theorem 4. *In the basis E_i of W:*

(a) *$n(Pe) = n(-e) = -M$ and $n(Te) = M$. A future-pointing frame of the same observer, $\mathrm{diag}(1, Q)\,e$, gives $n = Q^T M$: a rotation of M in the root of e, a rotation of $-M$ in the other.*

(b) *No $Q \in SO(3)$ has $QM = -M$. For a regular simplex M_D in $\mathbb{R}^D$, such a rotation exists if and only if D is even.*

(c) *The region $\mathrm{TetA} = \{x \in W : \max_a n_a \cdot x \le 1\}$ of e becomes $-\mathrm{TetA} = \{\max_a(-n_a \cdot x) \le 1\}$ for Pe, the all-flipped region TetB of Letter 1; together they form the stellated octahedron [8].*

Proof. (a) The frame vectors of Λe are $E\Lambda^{-1}$; for $\mathrm{diag}(1, Q)$ the spatial ones become $E'_j = \sum_i Q_{ji} E_i$, and $Q^T M = -(-Q^T)M$ with $\det(-Q^T) = -\det Q$. (b) $\mathrm{rank}\, M_D = D$, so $QM_D = -M_D$ forces $Q = -I_D$, of determinant $(-1)^D$. (c) $(-n_a) \cdot x = n_a \cdot (-x)$. □

Remark. The tetrahedron is congruent to its mirror image: a 90° rotation maps the vertex set of TetA onto that of TetB, but reorders the legs. The roots differ in the order of the legs and the region of W they occupy, not in shape. And the tetrahedron follows the root only for future-pointing frames: $-e$ carries TetB in the root of e, because it also reverses time.

Scope

Ingredient	Without it
$\det g \neq 0$	$c = 0$: a double root, not two
D odd	P keeps the root; for $D = 2$, $-M_2$ is M_2 turned by π
Future-pointing	$-e$ carries TetB in the same root
Ordered legs	As point sets TetB is a rotation of TetA

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